

ERNIE Part List

Carrying Case

Part list

No.	Part	Quantity	Available	On-site for Assembly ✓ (Yes/NO) ☐
1.	Case	1	Yes ▾	

Magnet

Part list

No.	Part	Quantity	Available	Added to Assembly ✓ / Not yet added to the Assembly ☐	Disassembled ✓ / Not yet Disassembled ☐
1.	Ring 42 (PLA)	2	Yes ▾		
2.	Ring 252	2	Yes ▾		
3.	Ring 288	2	Yes ▾		
4.	Ring 321	1	Yes ▾		
5.	Lids for Rings 42	2	Yes ▾		
6.	Lids for Rings 252	2	Yes ▾		
7.	Lids for Rings 288	2	Yes ▾		
8.	Lids for Ring 321	1	Yes ▾		
9.	180mm Brass Rods	8	Yes ▾		
10.	Ring Spacers	48	Yes ▾		
11.	M4 Nylon/ Brass Nuts	8	Yes ▾		
12.	M3 Nylon / Brass Screws	54	Yes ▾		

No.	Part	Quantity	Available	Added to Assembly ✓/ Not yet added to the Assembly ☐	Disassembled ✓/ Not yet Disassembled ☐
13.	Dome-Shaped Nuts	8	Yes ▾		
14.	Washers	16	Yes ▾		
15.	Neodymium Magnets	99	Yes ▾		
16.	Reference Magnets	1	Yes ▾		
17.	Heat Set inserts (Soldered into the rings)	54	Yes ▾		

Tools needed for Magnet Assembly

No.	Tool	Quantity	Available	Onsite for Assembly ✓/ Not onsite for Assembly ☐	Returned to box ✓/ Not returned to Box ☐
1.	Screw Drivers	4	Yes ▾		
2.	Spanners	2	Yes ▾		
3.	Compass	2	Yes ▾		
4.	Permanent Markers	4	Yes ▾		
5.	Vernier Caliper	1	Yes ▾		
6.	Ruler	1	Yes ▾		

Robot Assembly
Part list

No.	Part	Quantity	Available	Added to Assembly ✓/ Not yet added to the Assembly ☐	Disassembled ✓/ Not yet Disassembled ☐
1.	Magnet Rails	3	Yes ▾		
2.	Motor Drivers	3	Yes ▾		
3.	Arduino Uno	1	Yes ▾		
4.	3D Printed Bases	4	Yes ▾		
5.	Gauss Meter	1	Yes ▾		
6.	Screws	8	Yes ▾		
7.	Connecting Wires	25	Yes ▾		
8.	Power Supply	1	Yes ▾		
9.	Gauss Meter Probe Holder	1	Yes ▾		
10.	Computer	1	Yes ▾		
11.	Gauss meter probe calibrator	1	Yes ▾		
12.	Motor driver casing (3D printed)	1	Yes ▾		

Tools needed for Magnet Assembly

No.	Tool	Quantity	Available	Returned to box ✓/ Not returned to the box ☐
1.	Screw Drivers	4	Yes ▾	
2.	Spanners	2	Yes ▾	
3.	Allen Keys	4	Yes ▾	
4.	Masking Tape	1	Yes ▾	

Magnet Disassembly

No.	Part	Quantity	Available	On-site for Assembly ✓ (Yes/NO) ☐
1.	Magnet Ouster	1	Yes ▾	
2.	Disassembly Tray		Yes ▾	
3.	Disassembly Tray Cover		Yes ▾	